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# Predictive Analytics Using Historical Data
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import matplotlib.pyplot as plt

np.random.seed(42)
n = 300
data = pd.DataFrame({
    'Advertising_Spend': np.random.randint(5000, 50000, n),
    'Website_Visitors': np.random.randint(1000, 10000, n),
    'Discount': np.random.randint(0, 40, n)
})
data['Sales'] = (0.4 * data['Advertising_Spend'] + 15 * data['Website_Visitors']
                - 500 * data['Discount'] + np.random.randint(-5000, 5000, n))
X = data[['Advertising_Spend', 'Website_Visitors', 'Discount']]
y = data['Sales']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("MAE:", mean_absolute_error(y_test, y_pred))
print("MSE:", mean_squared_error(y_test, y_pred))
print("R2:", r2_score(y_test, y_pred))
new_data = pd.DataFrame({'Advertising_Spend': [25000], 'Website_Visitors': [7000], 'Discount': [10]})
print("Prediction:", model.predict(new_data)[0])
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Sales")
plt.ylabel("Predicted Sales")
plt.show()

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